DATA ANALYTICS PROJECT

Facebook Page Analytics Report

Business Intelligence Analysis | 16 SQL Queries | 7 Data Tables

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PRESENTATION AGENDA

01

Database Structure

Overview of the 7 tables and how data is organized

02

Basic Summary Statistics

Countries, cities, languages, reach and likes

03

Location & Fan Analysis

Top countries, penetration ratios, city growth potential

04

Language & Buying Power

English speakers, percentage split, revenue potential

05

Engagement Analysis

Best day and time of week to publish posts

06

Live SQL Demonstrations

3 queries run live with methodology explanation

DATABASE STRUCTURE

PopStats

Reference

150 countries

CountryCode, CountryName, Population, AverageIncome

FansPerLanguage

Language

41 languages

Date, Language, CountryCode, NumberOfFans

FansPerCountry

Location

Time-series

Date, CountryCode, NumberOfFans

GlobalPage

Performance

45 rows/day

Date, CountryCode, NewLikes, DailyPostsReach...

FansPerCity

Location

46 cities

Date, City, CountryCode, NumberOfFans

PostInsights

Engagement

Post-level

CreatedTime, EngagedFans, PostClicks, Reach...

FansPerGenderAge

Fan

Demographics

Date, Gender, AgeGroup, NumberOfFans

SECTION 1: BASIC SUMMARY STATISTICS

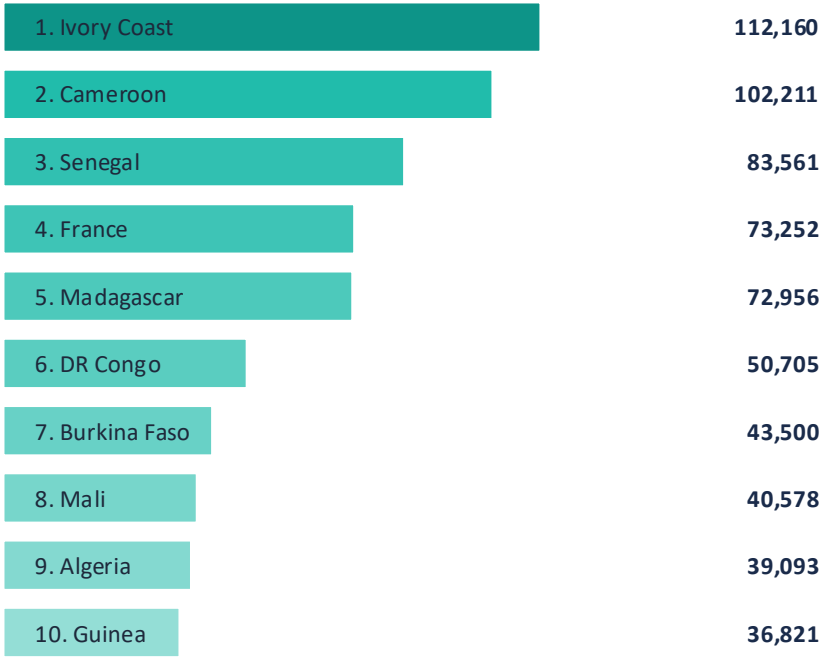
150	46	41	83.8M
Unique Countries	**Unique Cities**	**Languages**	**Avg Daily Reach**
PopStats table	FansPerCity table	FansPerLanguage table	Corrected methodology

⚠ KEY METHODOLOGY NOTE — Q4 Daily Average Reach

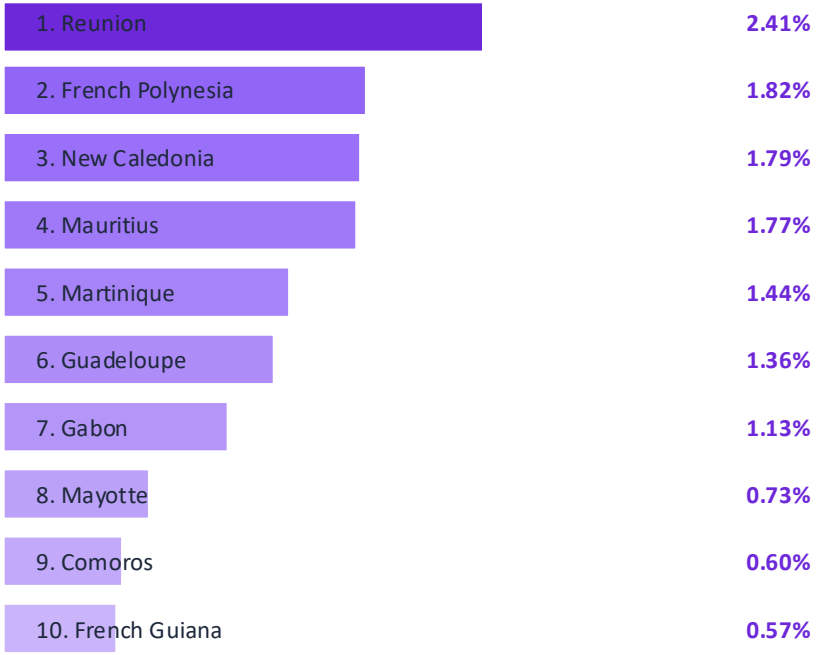
Original query: AVG(DailyPostsReach) directly → result: 1,862,816 ❌ (average per country per day, not global)
Corrected query: SUM per date first → then AVG of daily totals → result: 83,826,721 ✅ (true global daily reach)

SECTION 2: LOCATION ANALYSIS

Q6 — Top 10 Countries by Raw Fans



Q7 — Top 10 by Penetration Ratio (%)



KEY INSIGHT

Raw Fans ≠ Market Penetration

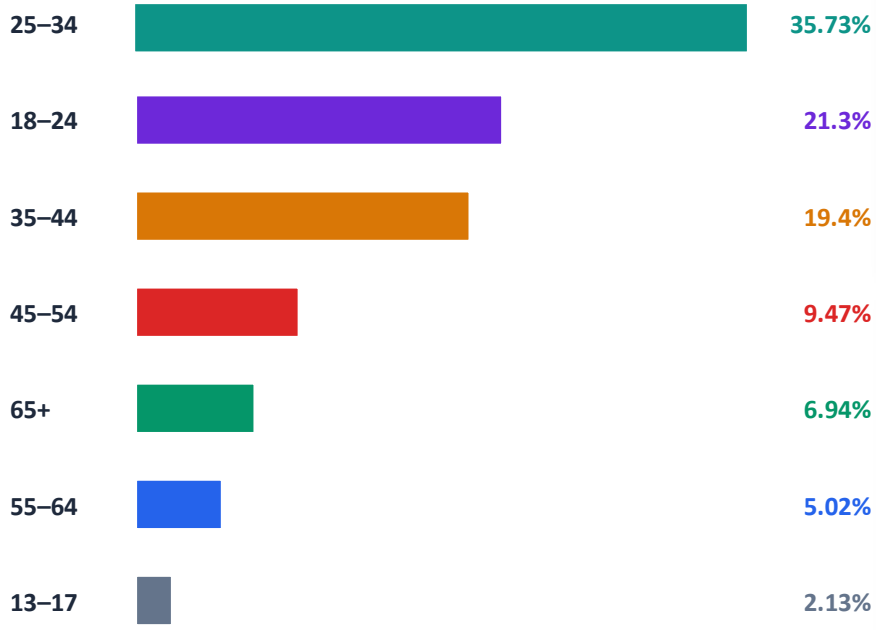
Ivory Coast leads with 112,160 fans but has a penetration ratio of only ~0.46% because its population is 24 million.

Reunion has only 20,885 fans but a 2.41% penetration ratio because its population is just 866,506.

💡 Penetration Ratio is a far more meaningful metric for identifying where the brand has genuine market presence relative to its available audience.

SECTION 3 & 4: FAN DEMOGRAPHICS & LANGUAGE ANALYSIS

Q9 — Fan Split by Age Group



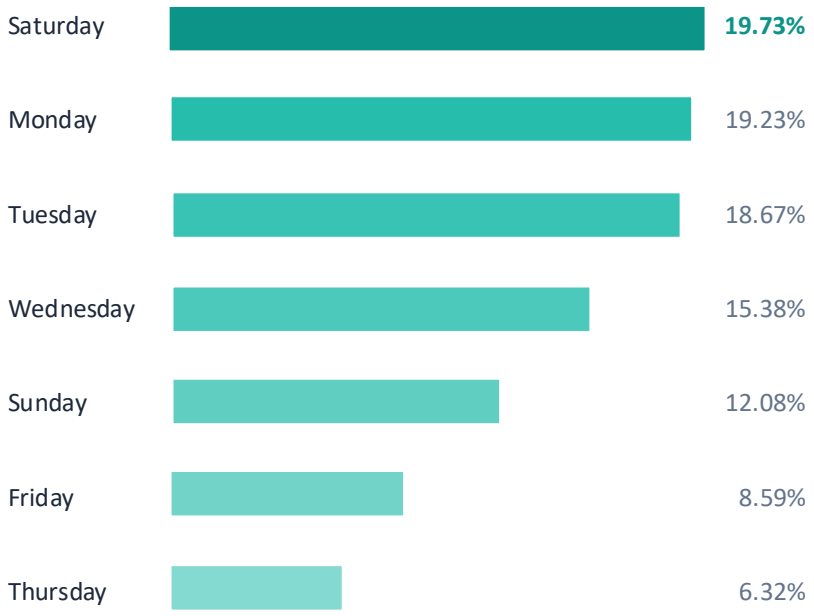
Q10 — Fan Split by Gender

Q11–Q13 — Language & Buying Power

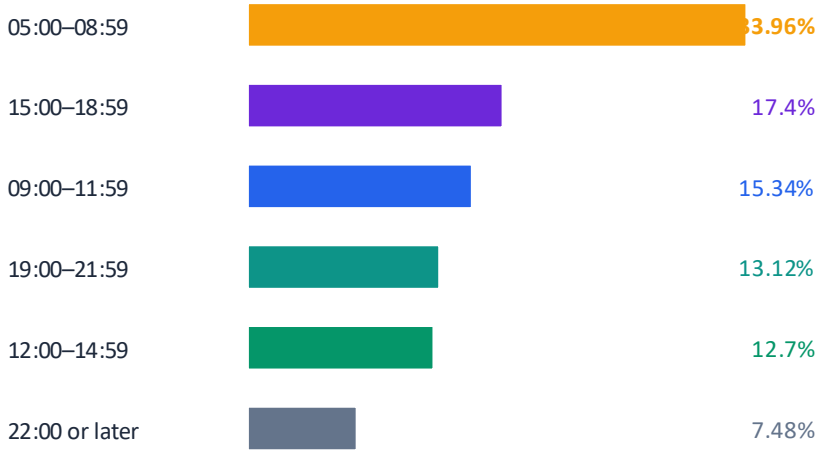
English-speaking fans	1,347,752
% of total fan base	5.08%
Potential buying power (US)	$5.46M

SECTION 5: FAN ENGAGEMENT ANALYSIS

Q14 — Engagement by Day of Week



Q15 — Engagement by Time of Day



⚠ Data Quality Caveat — Q14

Posts are almost entirely published on the 10th of each month. Day-of-week results reflect calendar coincidence, not genuine engagement patterns across all 7 days.

SECTION 6: LIVE SQL DEMONSTRATIONS

01

Q4 — Corrected Daily Average Reach

Why this query: Demonstrates deep table structure understanding and catching a methodology error

Result: **83,826,721.36 (vs incorrect 1,862,816)**

Key insight: Table has 45 rows/day — must SUM per date before AVG for true global figure

```
WITH globaldaily AS (
  SELECT date,
  SUM(dailypostsreach) AS globaldailyreach
  FROM globalpage GROUP BY date
)
SELECT ROUND(AVG(globaldailyreach), 2)
AS dailypostreach FROM globaldaily;
```

02

Q7 — Penetration Ratio

Why this query: Shows JOIN across two tables, proportional thinking, and self-correction of redundant SUM()

Result: **Top: Reunion 2.41%, French Polynesia 1.82%**

Key insight: Ivory Coast #1 in raw fans but absent from penetration ranking — small territories dominate

```
SELECT p.countryname,
ROUND((f.numberoffans*100.0)/p.population,2)
AS Penetration_Ratio,
f.numberoffans, p.population
FROM fanspercountry f
JOIN popstats p ON f.countrycode=p.countrycode
WHERE date=(SELECT MAX(date)
FROM fanspercountry)
ORDER BY Penetration_Ratio DESC LIMIT 10;
```

03

Q13 — Buying Power of English Speakers

Why this query: Connects data to a real business decision — estimating revenue potential

Result: **$5,464,504.63 potential buying power (US)**

Key insight: Upper-bound estimate; assumes average income and 100% subscription conversion rate

```
SELECT ROUND(
  SUM(f.NumberOfFans
  * p.AverageIncome * 0.0001), 2)
AS PotentialBuyingPower
FROM FansPerLanguage f
JOIN PopStats p
ON f.CountryCode=p.CountryCode
WHERE f.Language='en'
AND LOWER(p.CountryName)='united states';
```

KEY FINDINGS

Summary

Audience Base

150 countries, 46 cities, 41 languages. Predominantly French-speaking African nations — Ivory Coast leads with 112,160 fans.

Market Penetration

Small French overseas territories dominate penetration: Reunion 2.41%, French Polynesia 1.82%. More meaningful than raw fan count.

Demographics

56.46% female, 43.44% male. Largest age group is 25–34 (35.73%). Young female-skewing audience.

Language

Only 5.08% of fans are English speakers. Brand's core audience is non-English, aligning with French-African positioning.

Revenue Potential

$5.46M estimated potential buying power from English-speaking US fans. Upper-bound estimate with assumptions noted.

Best Time to Post

05:00–08:59 drives 33.96% of engagement. Day-of-week analysis is unreliable due to data distribution limitations.

Thank You

Ready for your questions, Alexandra.

Prepared by: Dhandapani Rathina Priya

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Queries demonstrated live: Q4 (Corrected Reach) • Q7 (Penetration Ratio) • Q13 (Buying Power)